



Peyjal Survekshan



***PROPOSAL FOR QUARTERLY SURVEY ASSESSMENT TO ENSURE
PREPAREDNESS OF BHOPAL MUNICIPAL CORPORATION FOR PEYJAL
SURVEKSHAN***

BY QUALITY COUNCIL OF INDIA

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ABOUT QUALITY COUNCIL OF INDIA

Quality Council of India (QCI) was established as a National body for Accreditation on recommendations of Expert Mission of EU after consultations in Inter-ministerial Task Force, Committee of Secretaries and Group of Ministers through a Cabinet decision in 1996. Accordingly, QCI was set up through a PPP model as an independent autonomous organization with the support of Government of India and the Indian Industry represented by the three premier industry associations,

- Associated Chambers of Commerce and Industry of India (ASSOCHAM),
- Confederation of Indian Industry (CII) and
- Federation of Indian Chambers of Commerce and Industry (FICCI).

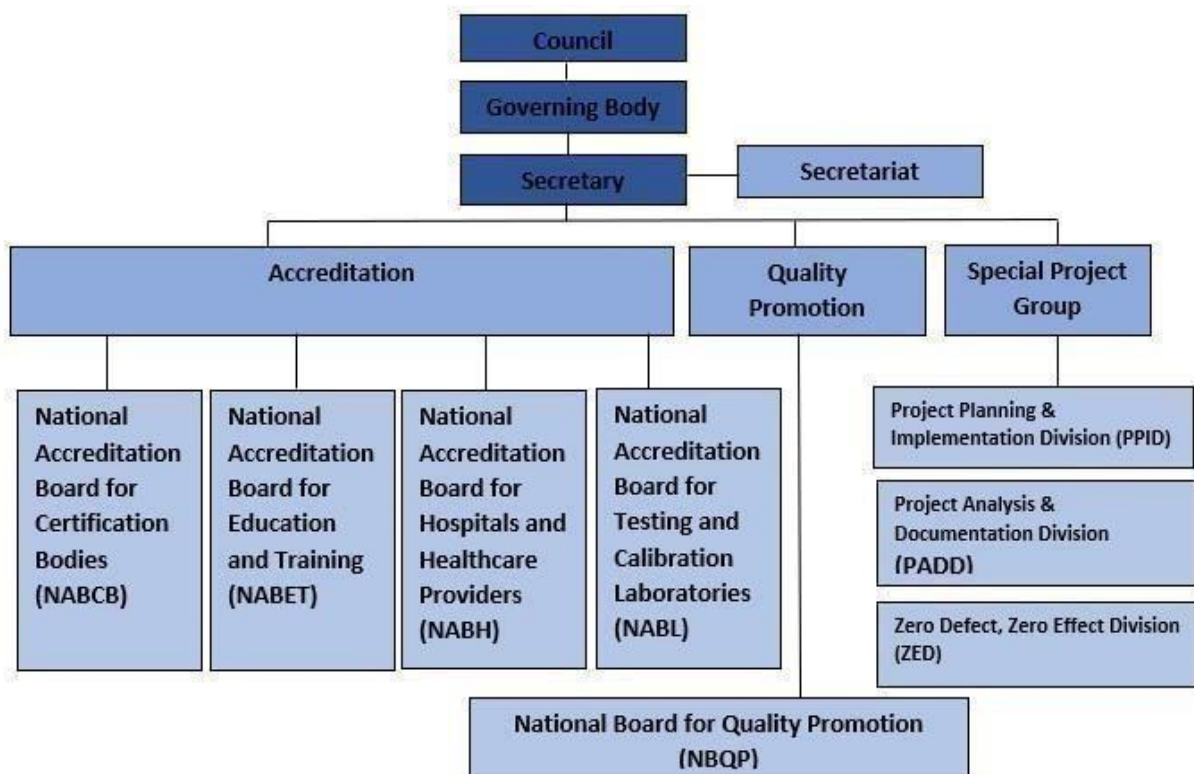
The Department of Industrial Policy and Promotion (DIPP), Ministry of Commerce and Industry is the nodal point for all matters related with QCI. The Chairman of QCI is appointed by the Hon'ble Prime Minister of India.

The Council was established to create a mechanism for third party assessment of products services and processes. In line with the international best practices, it was envisaged that the Council will play the role of the National Accreditation Body (NAB) as per existing eco-system of quality around the world and will ensure that the quality of life of citizens around the world is improved. It was therefore befitting that the Council was organized as an independent organization from both government and Industry and functions as an Independent Professional body with its own rules, powers and functions strictly as per the MoU of the society. QCI has a pool of resources to conduct the third-party verification exercise.

- a) We have signed an MoU with Indian Statistical Institute (ISI) to provide guidance on analysing, interpreting and presentation of statistically relevant data.
- b) We have on-boarded consultants with specialization of GIS, ground water management.
- c) Our senior directors and directors are technically qualified to lead the project with over 20 years of experience.

- d) The project Managers from our young professional program, constituting Project Planning and Implementation Division (PPID), are well versed with project management techniques and have executed large scale PAN India TPI (Third Party Inspection) projects in short deadlines.

The organization structure of Quality Council of India is depicted below:



QCI's PAST EXPERIENCE

Peyjal Survekshan (Pilot) 2021

The survekshan pilot study was a pre-cursor to a pan-India competitive platform to assess the present water and wastewater system, identify gaps, suggest best practices, and encourage the cities to move towards becoming water secure by adopting reduce, reuse and recycling principle. 8 cities were assessed for the pilot survey: Surat, Patiala, Bhubaneswar, Rohtak, Agra, Churu, Tumkur and Badlapur. The project aimed to assess water security status of Indian cities by analyzing and assessing water supply and distribution, wastewater collection and treatment, health of waterbodies and estimation of non-revenue water. The assessment process included citizen feedback, direct observation and water samples were collected from household taps and water bodies and tested at NABL accredited labs.

Piped Water Supply Scheme- Ministry of Drinking Water and Sanitation (2016-17)

The scheme aimed to recharge ground water through point recharge sources, basic treatment and re-use of grey-water for agriculture, de-silting minor irrigation tanks and rejuvenation of water bodies. Also, to provide piped water to villages where the quality of water is good. In villages with poor water quality, however, a trunk water supply system will be arranged. About 5600 villages spread across India were surveyed to assess the functionality, availability and sustainability of PWS schemes under NRDWP. A detailed report was then submitted to Ministry of Drinking Water and Sanitation.

Swachh Survekshan 2021- Ministry of Housing & Urban Affairs

Data captured using survey instruments/portal developed in consultation with MoHUA with real time tracking of collected data. Data collected from all the three sources shall be updated on the portal, especially Service Level Progress and Citizen's Voice.

SBM (ODF, ODF+, ODF++)–(2016-2021)- Ministry of Housing and Urban Affairs

The Ministry of Housing & Urban Affairs (MoHUA) has commissioned QCI to certify 4372 Urban Local Bodies of India as Open Defecation Free by October 2019 under the Swachh Bharat Mission. In a bid to step up the pace of the achievement, the MoHUA has introduced higher certifications ODF+ and ODF++. While ODF+ focuses on toilets with availability of water, maintenance and hygiene, ODF++ focuses on proper disposal and treatment of faecal sludge / septage and sewage, with no discharging and/or dumping of untreated faecal sludge/septage and sewage in drains, water bodies or open areas. It also focuses on waste water treatment through sanitation technologies like STPs and FSTPs. The survey is conducted with the help of mobile-based application which captured time stamped and geo tagged pictures for data transparency.

97 Ganga Towns– Ministry of Housing and Urban Affairs(2018)

The project envisaged the quality assessment of sanitation and cleanliness in and around the Ghats/rivers of Ganga towns spread across Indo-Gangetic planes. The parameters were broadly classified as Overall cleanliness, solid waste floating on the surface of river Ganga (usage of trash cleaners), solid waste management services, screening of nallahs and cleaning and removal of waste from nallah screens, tapping of nallahs to the STP, availability of MSW plant. The entire survey was conducted using a mobile application to minimize physical interventions and reduce the chances of error.

Swachh Survekshan (Urban) 2017 - Ministry of Housing and Urban Affairs

City Ranking Under Swachh Survekshan (Urban) 2017 surveyed the existing sanitation and solid waste management conditions in the city. A list of indicators pertaining to the practice of sweeping, collection, transportation, processing and disposal of solid waste, access to sanitation were surveyed. The survey includes collection of primary information, field and side visits to conduct interviews along with photo documentation of field condition. The information was captured over mobile application which captured real time, time stamped and geotagged pictures. Each inspection was supported by the team of professionals, guiding the inspectors captured real time, time stamped and geo tagged pictures. Each inspection was supported by the team of professionals, guiding the inspectors on-site.

Piped Water Supply Scheme- Ministry of Drinking Water and Sanitation (2016-17)

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Swachh Survekshan (Grameen)- Ministry of Drinking Water and Sanitation and FICCI (2016)

Assessment and ranking of 75 best performing districts (NSSO) in rural India on parameters of Swachh Bharat Mission-Grameen was conducted. A ranking of these districts was derived based on the scores obtained on various sanitation parameters. A total of 36 villages were covered in each district in the plains and 20 in hilly areas. A sample of 30 and 20 geographically spread households were surveyed in plain and hilly districts respectively. The survey was conducted with the help of mobile based application which captured time stamped and geotagged pictures. A detailed report was then submitted to the FICCI and the Ministry of Drinking Water & Sanitation as annexed.

INTRODUCTION

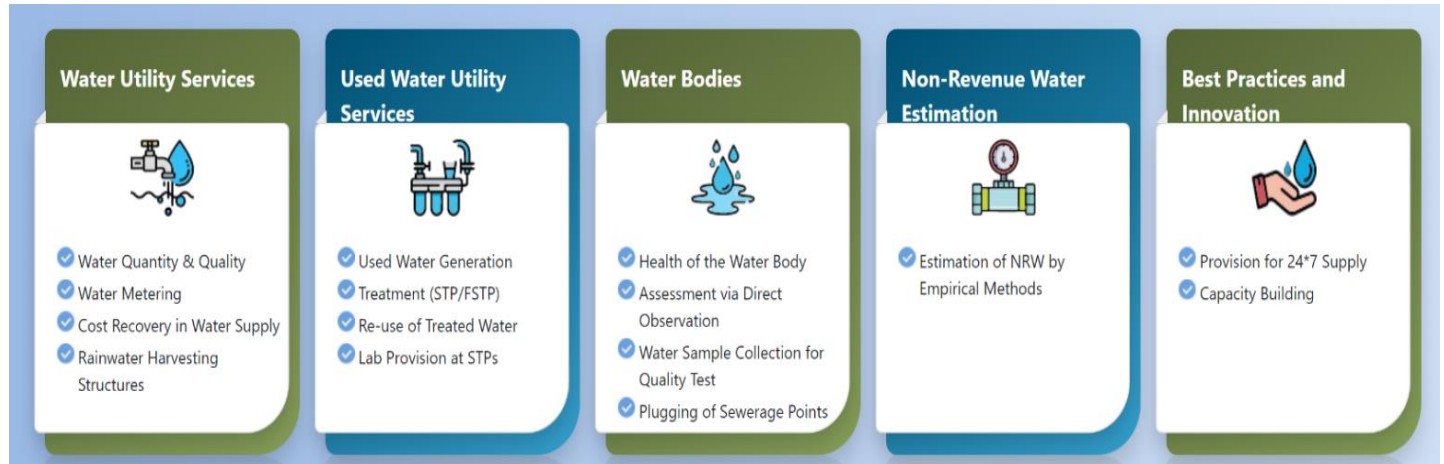
India has long faced the challenge of providing safe drinking water to over 700 million people in more than 1.5 million villages. In 1972, the government began to improve rural water supply, and in the mid-1980s the issue was declared a national priority. As a result, by 2011, 95 percent of India's rural population had access to some form of water supply infrastructure. In practice however many systems were no longer functional.

AMRUT 2.0 is a part of the Aatma Nirbhar Bharat initiative aimed at ensuring water security for cities and providing functional tap water connections to all households. The mission's primary objective is to offer tap water connections to households in all statutory towns through the implementation of 2.68 crores of new household tap connections.

Madhya Pradesh has 1.23 crore rural households, out of which 37.69 lakh (31%) have tap water supply in their homes. In 2020-21, Madhya Pradesh has provided 19.89 lakh tap water connections and has been one of the top performers in the country under Jal Jeevan Mission. In 2021-22, the State has planned for saturation of 7 districts and 22 lakh new tap water connections. Bhopal is a rapidly growing urban centre where the demand for clean water is escalating due to the increasing population and urbanization. Assessing the water health of the city of Bhopal is essential to ensure a safe and sustainable supply of water for the people.

Peyjal Survekshan (PJS) is launched in the AMRUT cities as an annual challenge to assess the compliance of service level benchmarks for quality, quantity, coverage of water supply, septage management and sewerage, extent of reuse & recycling of used water and conservation of water bodies within the city. It will showcase the best performing Urban Local Bodies (ULBs) in the country. In the spirit of cooperative federalism, the survekshan intends to foster healthy competition among ULBs by ranking them on the quality of drinking water, and motivate cities to implement the assigned tasks on time to improve their water health.

The core focus areas of Peyjal Survekshan as follows:



OBJECTIVES

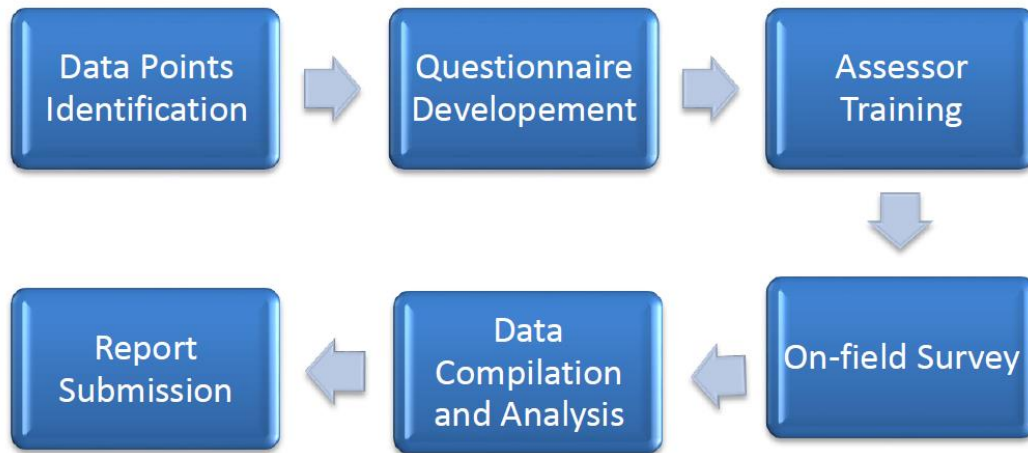
The objective of the proposal is to:

- 1.1 Assess compliance of
 - a) water supply and;
 - b) service level benchmarks
 with respect to coverage, quality, quantity, and user charges reforms.
- 1.2 Evaluate Bhopal MC's water supply management & innovative practices
- 1.3 Assess the operational efficiency of sewage and water treatment plants
- 1.4 Reduce non-revenue water (NRW) through district metered areas (DMAs) and training to check leakages
- 1.5 Analyze the rejuvenation of water bodies and wells
- 1.6 Monitor the collection, treatment, and reuse of treated used water

The evaluation of the Bhopal MC's water supply management the abovementioned indicators will help facilitate the preparedness of Bhopal Municipal Corporation (Bhopal MC), Madhya Pradesh in carrying out the Peyjal Survekshan seamlessly and allow it to compete with state and national level rankings. **4 assessments will be conducted in financial year (2023-2024)**

METHODOLOGY

The following methodology will be followed to carry out the assessment in the said locations:



1. Development of Survey Plan

On the basis of data points identified and the available survey tools, a comprehensive plan will be developed separately for different locations including residential and commercial areas with Bhopal MC for assessment of water supply management & innovative practices. The samples will be collected as per the approved sample plan.

2. Questionnaire Development

On the basis of data points identified and the available survey tools, a comprehensive questionnaire will be developed separately for different stakeholders including citizens, households, water bodies, water infrastructure, parks, etc. The questionnaire will be developed in consultation with Bhopal MC in accordance with prescribed water testing norms.

3. On-field assessor recruitment and training

QCI will leverage its pool of NABCB accredited assessors for the survey. The assessors with graduation degree and 2+ years of experience will be deployed for the survey. The assessors will be interviewed to check their communication skills and knowledge for the assessment. A test/evaluation will be conducted to ascertain that they are well-versed with the concept and assessment methodology.

4. Documentation from ULB

QCI will study the secondary literature and documentation of Bhopal MC to assess if the SLBs defined by MoHUA are being met through existing service delivery. The data collected from the cities will be cross verified with the data collected on field. It is also required that Bhopal MC grant QCI an authorization letter.

5. On-field Survey

The assessors will conduct field survey in the Bhopal using random sampling methodology. The sampling points will be as follows:

1. The assessors will visit select water sources, treatment infrastructure, RWH, infrastructure and green spaces at the city level.
2. Water testing:
 - Tap water samples will be collected from residential and slum areas and from WTP tested for IS 10500 parameters.
 - Treated wastewater collected from select STPs and tested as per CPHEEO standards.
 - Water body samples collected and tested as per CPCB standards.

The survey will have 2 main aspects:

5.1 Direct Observation

The assessors will visit the source sites, distribution reservoirs, water and used water treatment plant in the ULB Parks and RWH structures claimed by the ULB. They will check the physical health of the water body including fencing around the water body, solid waste in and around the water body, eutrophication, presence of animal dung, etc. Tap water samples will be collected from residential and slum areas and from WTP, treated used water collected from select STPs and tested as per CPCB norms, and water body samples collected and tested as per CPCB standards. The assessors will also interact with ULB officials, etc. All of the photographs and proof will be geo-tagged and the data will be displayed in real-time on the backend quality check portal. The aim will be to thoroughly understand the water management system in the Bhopal and help the MC prepare for the surveys.

5.2 Citizen Feedback

The CF assessor will travel the entire city to collect quantitative inputs on aspects of water quantity, quality, water meter availability, bill payments, grievance redressal, awareness on water conservation etc., from individuals/ commercial vendors/ industrial units/ schools/ hospitals, etc. The questions will be framed for all stakeholders.

6. Compilation and Analysis of data received

The data collected from the field, post the quality check, will be analyzed critically. The outputs to all desired data points will be collated in the form of a report. Gaps and recommendations will be highlighted for operating more efficiently and effectively in the water management and conservation process.

7. Support Required

We would need support from Bhopal MC on the following aspects-

1. To get authorizations from Bhopal MC to carry out survey in boundary of Bhopal MC.
2. Any information that is required time to time to frame the questionnaire or documentation.
3. Appointment of nodal officer from the Bhopal MC to facilitate the process.
4. Any other support as and when may be required.

QUALITY CONTROL MECHANISM

Quality Control Mechanism Training

To ensure the consistency of the survey the assessors will be trained in detail on the concepts and on the nitty-gritties of the project. The questionnaires involving the corresponding photographs to the questions will be explained in detail. The assessors will also be trained on the usability of the technical know-how of the QCI data collection mobile application.

24*7 War room

A 24*7 war room will be set up at Delhi office of QCI. The team will regularly conduct the real time quality check of the collected data. The back-end team of QCI shall use statistic tools like correlation, regression, thematic map using software like Tableau, Microsoft Excel and GIS. The team will draw inferences from the collected data and will present the same to client.

Data Monitoring and Control

A dedicated control room shall be setup at QCI to monitor the daily progress, consistency and quality of the survey. The survey results shall be monitored on a real time basis and the queries of the field assessors will be resolved promptly by the team.

The data collected by on field assessors undergoes a thorough and rigorous three-layer quality checks:

Level 1: Trained field assessor collects the data on-ground and uploads it on a real time basis through mobile application.

Level 2: Quality check coordinator (stationed at QCI headquarters) who is mapped to a field assessor shall monitor the survey and validate the information received at the back end.

Level 3: A senior QCI team member (experts) at the control room re-evaluates the field data collected.

EXPECTED OUTCOMES

- Assessing the compliance of service level benchmarks for water supply, sewerage & septage management, extent of reuse & recycle of used water and conservation of water bodies in Bhopal.
- Understanding of citizen outreach regarding the water management. This can be a starting point to devise a plan at a later stage.
- Document any innovative practices regarding water supply management in Bhopal.
- Instil healthy competition among cities and while being a monitoring tool and Mission accelerator.
- Corrective measures for non-functional aspects of water supply, parks and water bodies will be highlighted.

TIMELINE

Month	Task to be completed
February 2023	Field Assessment
March 2023	Submission of Report
June 2023	Field Assessment
July 2023	Submission of Report
October 2023	Field Assessment
November 2023	Submission of Report
January 2024	Field Assessment
February 2024	Submission of Report

FINANCES AND MANPOWER

Particulars	Cost	Man-Month	Total (in Rupees)
Assessor	5000	75	375000
QCI Resource cost	60000	4	240000
Report Design	25000		25000
Sub-total			640000
QCI admin cost			76800
Pre-tax total amount			716800

**All the quoted prices are for per-quarter basis.*

*** 4 assessments will be conducted in financial year (2023-2024).*

PAYMENT SCHEDULE

S. No.	Target or milestones achieved	Percentage of contract value
1	Advance (per assessment cost)	50%
2	Final report submission	50%

GENERAL TERMS

QCI will conduct all these tasks with the highest standard of performance.

• **AMENDMENT OF THE AGREEMENT:** No amendment or modification of this agreement shall be valid unless it is made in writing jointly by both parties or by the authorized representatives and specially stating the reason for amendment to this agreement. The modification/changes shall be effective from the date of which they are made/executed unless otherwise agreed to.

• **FORCE MAJEURE:** The parties shall ensure due compliance with the terms of this agreement. However, no party shall be liable for any claim for any loss or damage whatsoever arising out of failure to carry out the terms of the agreement to the extent such a failure is due to force majeure events such as fire, rebellion, mutiny, civil commotion, riots, strike, lockout, force of nature, accident, act of God, or any other reason beyond the control of concerned party.

• **ARBITRATION:** The disputes, if arising, shall be settled by arbitration in accordance with the procedure prescribed in arbitration and Conciliation Act of India, 1966 or any statutory modification or re-enactment thereof and the rules made there under and for the time being in force. Proceedings shall, unless otherwise agreed by the parties, be held in Delhi, India.

• **NOTICE:** All notices permitted or required to be given under the provisions of this agreement shall be deemed to be sufficiently given if written and delivered personally or sent to the other party by telefax or certified or registered air mail or email and addressed to the registered office of the other party as follows or to such other party address as the party may give notice in writing.

Unless otherwise provided here in all such notices shall be deemed to have been given as of the date they are personally delivered or transmitted by telefax or email or deposited in the mail.